

Department of Computer Engineering, Punjabi University, Patiala
Subject: Object Oriented Programming
Class: 3rd Semester

MM. 15

MST-I

(Section-A) Answer all the questions

1. What is the output of the following program?

```
#include<iostream>
using namespace std;
class abc {
    void f();
    void g();
    int x;
};
main() { cout<<sizeof(abc)<<endl; }
```

2. What is encapsulation? How do we implement in C++?
3. What is the difference between new() and malloc()?
4. Define Friend Function.
5. What is the role of Scope Resolution operator? Give an example.

(Section-B) Answer any two questions

6. Write a Program to concatenate two strings using Operator Overloading of '+' operator.
7. Write a program to calculate area of circle and rectangle using the concept of constructor overloading.
8. Explain the basic concepts of Object Oriented Programming. Elaborate them in C++ syntax.

MST-I
CSEB1205T: Object Oriented Programming using C++
04/2022

Max. Marks: 15

Time Allowed: 60 Minutes

Note: Students must mention their class group on the paper.

N.B.: Section A is *compulsory*. Attempt *any two* questions from Section-B.

SECTION-A

- Q1
- i) Write four operators that cannot be overloaded.
 - ii) Differentiate between a structure and a class.
 - iii) Explain the concept of memory leak.
 - iv) Write the merits of OOP approach.
 - v) What is the benefit of using inline functions?

1
1
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SECTION-B

- Q2. What is a constructor? Explain various types of constructors.
- Q3. What is operator overloading? Write a program to overload a binary operator.
- Q4. Explain how memory allocation for objects is done in OOP.

5
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5

Object Oriented Programming using C++
Department of Computer Science and Engineering
Punjabi University, Patiala

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Date: 07-03-2025

Time: 60 Min

Section -A (Attempt all questions)

- | | |
|---|---|
| 1) What is a friend class. Write an example. | 2 |
| 2) What are static objects? Why do we need it. | 1 |
| 3) Write a function to demonstrate an array of objects. | 2 |

Section -B (Attempt any two questions)

- | | |
|---|---|
| 4) Write program in C++ to demonstrate various types of constructors i.e. Default, Parametrized and Copy constructor. | 5 |
| 5) Write program in C++ to demonstrate binary/ unary operator overloading. | 5 |
| 6) What is function overloading in C++. Write its program in C++. | 5 |

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
PUNJABI UNIVERSITY, PATIALA
Object Oriented Programming System

MST-1

Section-A (Do All)

- Q1) What is data hiding? How is it achieved in C++? (2)
- Q2) What do you understand by array of class object? Explain with example (1)
- Q3) What is destructor? How is it defined? (1)
- Q4) What is do you mean by memory allocation of member function? (1)

Section-B (Any two)

- Q5) W.A.P to illustrate the use of default, parameterized and copy constructor. (5)
- Q6) Write a program in C++ to input and display addition of two numbers, illustrating the concept of operator overloading. (5)
- Q7) Explain the features of oops and explain program development code (5)

Teacher:

Er. Anantdeep kaur